

CSA1711 - Artificial Intelligence

CO3 Assessment Tool - 3

Resolution Algorithm Implementation - Answer Script

Question 1: Rain and Wet Ground

Knowledge Base: 1. If it rains, then the ground becomes wet. 2. It is raining.

Goal: Prove that the ground is wet using the Resolution Algorithm.

1. Propositional Logic Representation

Let R = 'It rains' W = 'The ground is wet'
KB: $R \rightarrow W$ (Rule)
 R (Fact: it is raining)
Goal: W

2. Convert Statements to CNF

$R \rightarrow W \equiv \neg R \vee W$ (eliminate implication; already a single clause, no further steps needed)
Fact: R (already a unit clause)

3. Negate the Goal

$\neg W$ (added to the clause set for refutation)

4. Resolution Algorithm — Step by Step

Step	Clause A	Clause B	Resolvent
1	R	$\neg R \vee W$	W
2	W (step 1)	$\neg W$ (negated goal)	Empty clause $\square$

5. Conclusion

R $\neg R \vee W$
 \
 W $\neg W$
 \
 Empty clause $\square$

Result: The empty clause $\square$ is derived — a contradiction. Therefore the negated goal $\neg W$ is false, which proves the original goal: 'The ground is wet' is TRUE.

Question 2: Student Assignment Submission

Knowledge Base: 1. If a student submits the assignment, then the student receives internal marks. 2. Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks using the Resolution Algorithm.

1. Propositional Logic Expressions

Let S = 'Rahul submitted the assignment' M = 'Rahul receives internal marks'

KB: $S \rightarrow M$ (Rule)

S (Fact: Rahul submitted the assignment)

Goal: M

2. Convert Premises to CNF

$S \rightarrow M \equiv \neg S \vee M$ (eliminate implication)

Fact: S (unit clause)

3. Negate the Goal

$\neg M$

4. Apply the Resolution Algorithm

Step	Clause A	Clause B	Resolvent
1	S	$\neg S \vee M$	M
2	M (step 1)	$\neg M$ (negated goal)	Empty clause $\square$

5. Conclusion

Result: The empty clause $\square$ is derived, so the goal is **PROVED** — 'Rahul receives internal marks' is **TRUE**.

Question 3: Library Membership

Knowledge Base: 1. If a person is a library member, then the person can borrow books. 2. Priya is a library member.

Goal: Prove that Priya can borrow books using the Resolution Algorithm.

1. Propositional Logic Representation

Let L = 'Priya is a library member' B = 'Priya can borrow books'

KB: $L \rightarrow B$ (Rule)

L (Fact: Priya is a library member)

Goal: B

2. Convert Each Statement to CNF

$L \rightarrow B \equiv \neg L \vee B$ (eliminate implication)

Fact: L (unit clause)

3. Negate the Goal

$\neg B$

4. Resolution — Until No New Clauses Can Be Generated

Step	Clause A	Clause B	Resolvent
1	L	$\neg L \vee B$	B
2	B (step 1)	$\neg B$ (negated goal)	Empty clause $\square$

After step 2, the empty clause is produced; no further clauses remain to resolve, so the algorithm terminates.

5. Final Conclusion

Result: Contradiction (empty clause $\square$) derived — the goal 'Priya can borrow books' is **PROVED TRUE**.

Question 4: Placement Eligibility

Knowledge Base: 1. If a student clears the aptitude test, then the student is eligible for placement. 2. Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement using the Resolution Algorithm.

1. Propositional Logic Conversion

Let A = 'Arun cleared the aptitude test' E = 'Arun is eligible for placement'
KB: $A \rightarrow E$ (Rule)
 A (Fact: Arun cleared the aptitude test)
Goal: E

2. CNF Clauses

$A \rightarrow E \equiv \neg A \vee E$ (eliminate implication)
Fact: A (unit clause)

3. Negate the Goal

$\neg E$

4. Resolution Algorithm — Step by Step

Step	Clause A	Clause B	Resolvent
1	A	$\neg A \vee E$	E
2	E (step 1)	$\neg E$ (negated goal)	Empty clause $\square$

5. Resolution Tree and Conclusion

A $\neg A \vee E$
 \
 E $\neg E$ (negated goal)
 \
 Empty clause $\square$

Result: The empty clause $\square$ is derived — the negation of the goal leads to a contradiction. Hence 'Arun is eligible for placement' is PROVED TRUE.

Question 5: Access Control System

Knowledge Base: 1. If a user enters the correct password, then the user is authenticated. 2. If a user is authenticated, then the user is granted access. 3. The user entered the correct password.

Goal: Prove that the user is granted access using the Resolution Algorithm.

1. Propositional Logic Representation

Let P = 'User enters the correct password'
 Au = 'User is authenticated'
 Ac = 'User is granted access'
KB: $P \rightarrow Au$ (Rule 1)
 $Au \rightarrow Ac$ (Rule 2)
 P (Fact: user entered the correct password)
Goal: Ac

2. Convert All Implications to CNF

$P \rightarrow Au \quad \equiv \quad \neg P \vee Au \quad (\text{eliminate implication})$
 $Au \rightarrow Ac \quad \equiv \quad \neg Au \vee Ac \quad (\text{eliminate implication})$
Fact: P (unit clause)

3. Negate the Goal

$\neg Ac$

4. Apply Resolution to Derive the Empty Clause

Step	Clause A	Clause B	Resolvent
1	P	$\neg P \vee Au$	Au
2	Au (step 1)	$\neg Au \vee Ac$	Ac
3	Ac (step 2)	$\neg Ac$ (negated goal)	Empty clause $\square$

5. Explanation of Each Step and Final Conclusion

Step 1: The fact P (correct password entered) resolves with the clause $\neg P \vee Au$, cancelling P and $\neg P$, leaving the derived fact Au (user is authenticated).

Step 2: The derived fact Au resolves with the clause $\neg Au \vee Ac$, cancelling Au and $\neg Au$, leaving the derived fact Ac (user is granted access).

Step 3: The derived fact Ac resolves with the negated goal $\neg Ac$, cancelling both literals completely and producing the empty clause $\square$ — a contradiction.

Final Conclusion: Since assuming the negation of the goal ($\neg Ac$) leads to a contradiction, the original goal is proved: 'The user is granted access' is TRUE.